Lab Sheet: 3

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Objectives

- To understand the use of COMMIT and ROLLBACK commands in transaction control.
- To use ORDER BY and GROUP BY clauses in SQL queries.
- To understand DCL commands (GRANT, REVOKE) and the use of SAVEPOINT.
- To understand and implement JOIN operations in SQL.

Instructions

Open your preferred Database Management System (DBMS) tool.

1. Create the required tables: TEACHER, SALES, DEPARTMENTS, and EMPLOYEES.
2. Insert suitable records into each table.
3. Execute the SQL commands for transaction control, privilege control, grouping, sorting, and joins.
4. Observe the output after each operation such as ROLLBACK, COMMIT, SAVEPOINT, GRANT, and REVOKE.

DDL Questions (Data Definition Language)

1. Create the **TEACHER** table with the following attributes: (Name, DeptNo, Date_of_Joining, DeptName, Location, Salary).
2. Create the **SALES** table with the following attributes: (Sales_No, Salesname, Branch, Salesamount, DOB).
3. Create the **DEPARTMENTS** table with the following attributes: (dept_no, dept_name, dept_location).
4. Create the **EMPLOYEES** table with the following attributes: (emp_id, emp_name, emp_salary, dept_no).

DML Questions (Data Manipulation Language)

From TEACHER Table

1. Insert five records into the **TEACHER** table.
2. Give an increment of 25% salary to teachers working in the Mathematics department.
3. Give an increment of 15% salary to teachers working in the Commerce department.

From SALES Table

1. Insert five records into the **SALES** table.
2. Calculate the total sales amount in each branch.
3. Calculate the average sales amount in each branch.
4. Display the Salesname and DOB of salesmen born in the month of December with date format like *21-Dec-09*.
5. Display the Salesname and DOB of salesmen in alphabetical order of the month.

From JOIN Operations

1. Display employee details and department details where the department numbers are the same in both tables.
2. Display the employee name and department name using a LEFT OUTER JOIN.
3. Display the employee name and department name using a RIGHT OUTER JOIN.
4. Display the details of employees who draw a salary greater than the average salary.

TCL Questions (Transaction Control Language)

1. Perform the **ROLLBACK** command on the TEACHER table.
2. Perform the **COMMIT** command on the TEACHER table.
3. Write a query to implement a **SAVEPOINT**.

DCL Questions (Data Control Language)

1. Write a query to **grant all privileges** on the EMPLOYEES table to DEPARTMENTS.
2. Write a query to **grant selected privileges** on the EMPLOYEES table to DEPARTMENTS.
3. Write a query to **revoke all privileges** on the EMPLOYEES table from DEPARTMENTS.
4. Write a query to **revoke selected privileges** on the EMPLOYEES table from DEPARTMENTS.

```
CREATE DATABASE DBMS_Lab3;

USE DBMS_Lab3;


CREATE TABLE TEACHER (

   Name VARCHAR(50),

   DeptNo INT,

   Date_of_Joining DATE,

   DeptName VARCHAR(50),

   Location VARCHAR(50),

   Salary DECIMAL(10,2)

);


CREATE TABLE SALES (

   Sales_No INT PRIMARY KEY,

   Salesname VARCHAR(50),

   Branch VARCHAR(50),

   Salesamount DECIMAL(10,2),

   DOB DATE

);

CREATE TABLE DEPARTMENTS (

   dept_no   INT   PRIMARY   KEY,

dept_name          VARCHAR(50),

dept_location VARCHAR(50)

);
```

```
CREATE TABLE EMPLOYEES (     emp_id
INT PRIMARY KEY,     emp_name
VARCHAR(50),     emp_salary
DECIMAL(10,2),     dept_no INT,
   FOREIGN KEY (dept_no) REFERENCES DEPARTMENTS(dept_no)
);

INSERT INTO TEACHER VALUES
('Ravi', 101, '2015-06-12', 'Mathematics', 'Mumbai', 50000),
('Anita', 102, '2016-07-15', 'Commerce', 'Delhi', 45000),
('Suresh', 101, '2018-03-20', 'Mathematics', 'Chennai', 48000),
('Priya', 103, '2017-01-10', 'Science', 'Bangalore', 47000),
('Kiran', 102, '2019-09-05', 'Commerce', 'Hyderabad', 43000);
SET SQL_SAFE_UPDATES = 0;
UPDATE TEACHER
SET Salary = Salary + (Salary * 0.25)
WHERE DeptName = 'Mathematics';

SET SQL_SAFE_UPDATES = 1;

UPDATE TEACHER
SET Salary = Salary + (Salary * 0.15)
WHERE DeptName = 'Commerce';

INSERT INTO SALES VALUES
(1, 'Amit', 'Mumbai', 20000, '1995-12-21'),
(2, 'Neha', 'Delhi', 15000, '1998-05-14'),
(3, 'Rohan', 'Mumbai', 18000, '1996-12-09'),
```

(4, 'Sneha', 'Chennai', 22000, '1997-08-19'),

(5, 'Kunal', 'Delhi', 17000, '1994-12-30');

```
SELECT Branch, SUM(Salesamount) AS Total_Sales
FROM SALES
GROUP BY Branch;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Branch	Total_Sales
Mumbai	38000.00
Delhi	32000.00
Chennai	22000.00

```
SELECT Branch, AVG(Salesamount) AS Average_Sales
FROM SALES
GROUP BY Branch;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Branch	Average_Sales
Mumbai	19000.000000
Delhi	16000.000000
Chennai	22000.000000

```
SELECT Salesname,
    DATE_FORMAT(DOB, '%d-%b-%y') AS DOB
FROM SALES
WHERE MONTH(DOB) = 12;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Salesname	DOB
Amit	21-Dec-95
Rohan	09-Dec-96
Kunal	30-Dec-94

```
SELECT Salesname,
    DATE_FORMAT(DOB, '%d-%b-%y') AS DOB
FROM SALES
ORDER BY MONTHNAME(DOB);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Salesname	DOB
Sneha	19-Aug-97
Amit	21-Dec-95
Rohan	09-Dec-96
Kunal	30-Dec-94
Neha	14-May-98

```
INSERT INTO DEPARTMENTS VALUES
(1, 'HR', 'Mumbai'),
(2, 'Finance', 'Delhi'),
(3, 'IT', 'Bangalore');

INSERT INTO EMPLOYEES VALUES
(101, 'Arjun', 50000, 1),
(102, 'Meera', 60000, 2),
(103, 'Vikram', 55000, 3),
(104, 'Pooja', 45000, 2);

SELECT *
```

```
FROM EMPLOYEES E
JOIN DEPARTMENTS D
ON E.dept_no = D.dept_no;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

emp_id	emp_name	emp_salary	dept_no	dept_no	dept_name	dept_location
101	Arjun	50000.00	1	1	HR	Mumbai
102	Meera	60000.00	2	2	Finance	Delhi
104	Pooja	45000.00	2	2	Finance	Delhi
103	Vikram	55000.00	3	3	IT	Bangalore

```
SELECT E.emp_name, D.dept_name
FROM EMPLOYEES E
LEFT JOIN DEPARTMENTS D
ON E.dept_no = D.dept_no;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

emp_name	dept_name
Arjun	HR
Meera	Finance
Vikram	IT
Pooja	Finance

```
SELECT E.emp_name, D.dept_name
FROM EMPLOYEES E
RIGHT JOIN DEPARTMENTS D
ON E.dept_no = D.dept_no;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

emp_name	dept_name
Arjun	HR
Meera	Finance
Pooja	Finance
Vikram	IT

SELECT *

FROM EMPLOYEES

WHERE emp_salary > (SELECT AVG(emp_salary) FROM EMPLOYEES);

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

emp_id	emp_name	emp_salary	dept_no
102	Meera	60000.00	2
103	Vikram	55000.00	3
NULL	NULL	NULL	NULL